Banking Fragility and the Capitalization Dilemma[1]

Online Appendix

Contents

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A Empirical robustness and additional figures

This appendix section provides additional empirical evidence for the results in Section 2 of the main text. We report the annual IRF omitted from the main text, the three non-main productivity-shock IRF variants, and detailed baseline coefficient paths.

A.1 Additional IRFs

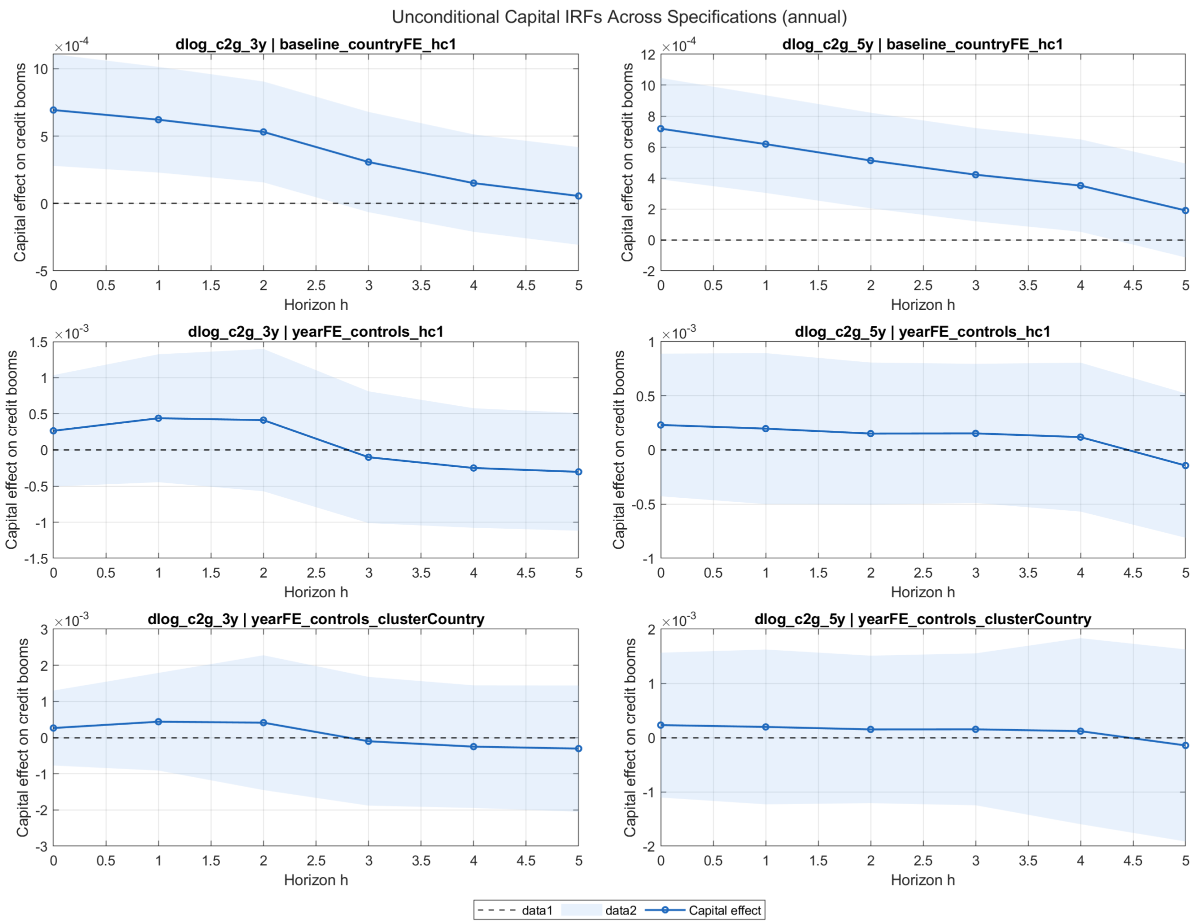

Figure A.1: Annual IRF: response of credit-boom measures to lagged capital ratios

Notes: Annual-horizon counterpart to the cumulative IRF shown in the main text.

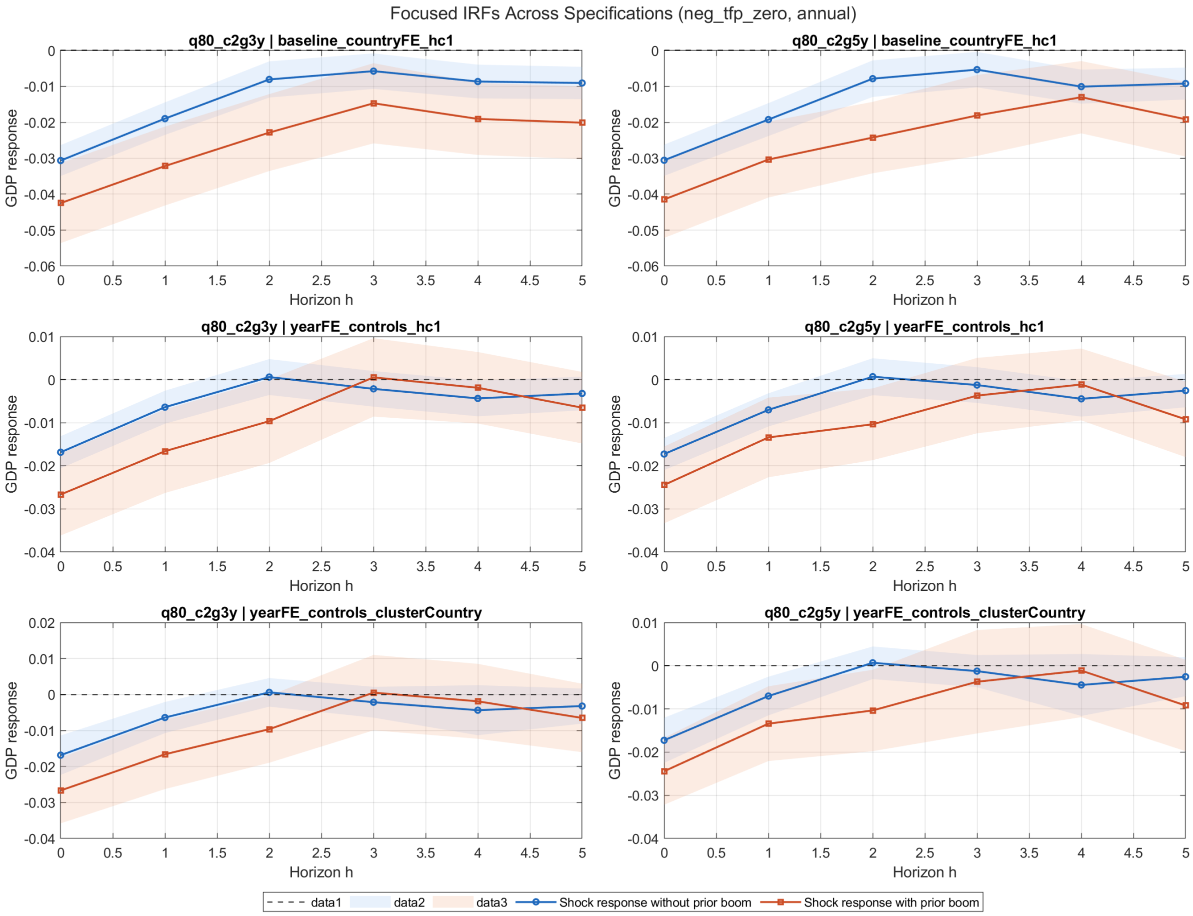

Figure A.2: Annual IRF: output response interaction (negative TFP defined as below zero)

Notes: Annual-horizon interaction estimates between negative TFP shocks and pre-shock credit-boom state.

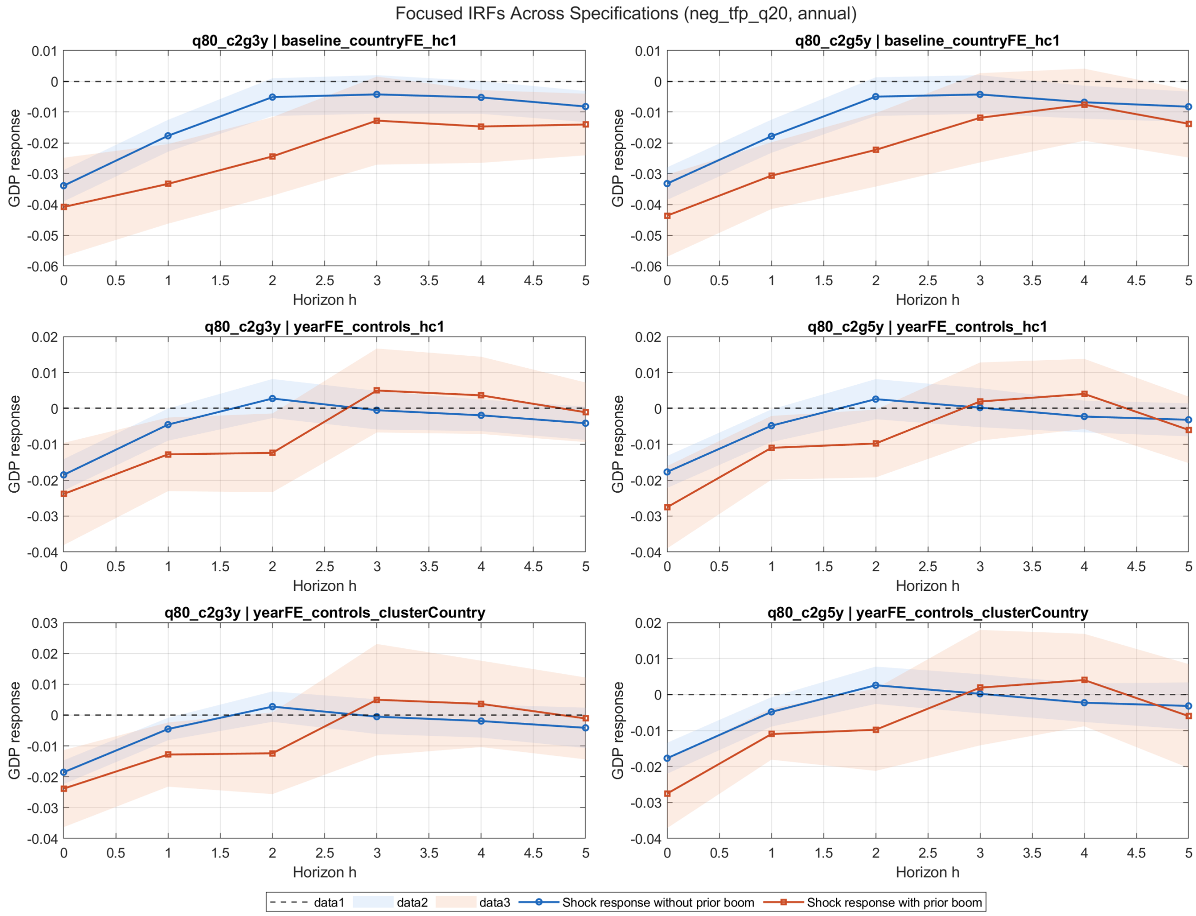

Figure A.3: Annual IRF: output response interaction (negative TFP defined as bottom quintile)

Notes: Alternative shock definition using the lower 20th percentile of TFP growth.

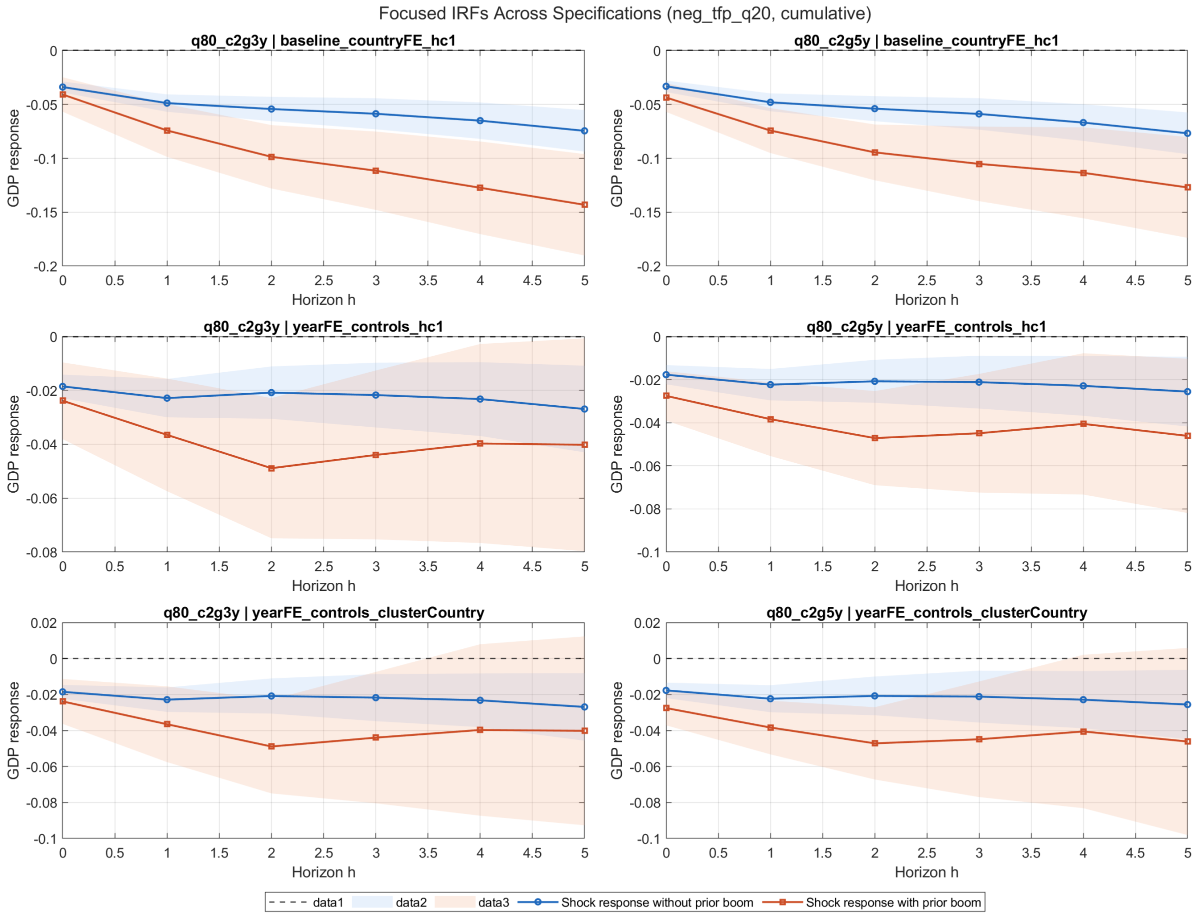

Figure A.4: Cumulative IRF: output response interaction (negative TFP defined as bottom quintile)

Notes: Cumulative-horizon counterpart to Figure A.3.

A.2 Detailed robustness tables

Table A.1: Baseline cumulative LP coefficients: credit boom on lagged capital ratio

Horizon	Coefficient	Std. error	p-value	95% CI low	95% CI high
0	0.00027	0.00049	0.595	-0.00077	0.00130
1	0.00067	0.00112	0.554	-0.00169	0.00304
2	0.00103	0.00201	0.617	-0.00324	0.00529
3	0.00089	0.00282	0.755	-0.00508	0.00686
4	0.00061	0.00356	0.866	-0.00694	0.00817
5	0.00034	0.00433	0.938	-0.00884	0.00952

Notes: Baseline specification with year fixed effects, controls, and country-clustered inference. Outcome is cumulative credit-boom response.

Table A.2: Baseline cumulative LP interaction coefficients: negative TFP $\times$ prior credit boom

Horizon	Interaction	Std. error	p-value	95% CI low	95% CI high
0	-0.01184	0.00613	0.054	-0.02387	0.00019
1	-0.02707	0.01023	0.008	-0.04714	-0.00700
2	-0.04330	0.01334	0.001	-0.06948	-0.01712
3	-0.05399	0.01634	0.001	-0.08605	-0.02194
4	-0.06448	0.01929	0.001	-0.10232	-0.02664
5	-0.07535	0.02165	0.001	-0.11783	-0.03286

Notes: Baseline specification with country fixed effects and robust inference. Shock definition uses negative TFP realizations below zero and a high-credit-boom pre-shock state.

B Model derivations

This appendix provides detailed derivations of the model's optimality conditions and equilibrium characterization.

B.1 Bank's problem

The bank maximizes its franchise value:

$$\begin{aligned} J(n;X) = \max_{n'}\ & \pi(n;X) + n - n' \\ & + (1-\mathcal{H}(X))\,\mathbb{E}\,q(X,X')\,J(n';X') \\ & + \mathcal{H}(X)\,\mathbb{E}\,q(X,X')\,J\big((1-\varphi)n';X'\big) \end{aligned} \tag{1}$$

where bank profit is:

$$\pi(n;X) = \max_{l}\ \Phi(l)(R^K(X)-1) - (R^B(X)-1)(l-n) \tag{2}$$

subject to the leverage constraint $b := l - n \leq \phi_0 + \phi_1 n$.

Intratemporal lending FOC. The first-order condition for optimal lending l is:

$$\Phi'(l)(R^K(X)-1) = (R^B(X)-1) + \lambda \tag{3}$$

where $\lambda \geq 0$ is the multiplier on the leverage constraint, with complementary slackness:

$$\lambda \geq 0, \quad l-n \leq \phi_0 + \phi_1 n, \quad \lambda\,[\phi_0 + \phi_1 n - (l-n)] = 0 \tag{4}$$

With $\Phi(l) = \Phi_A l^{\Phi_B}$, we have $\Phi'(l) = \Phi_A \Phi_B l^{\Phi_B - 1}$. Solving for l:

$$l = \left(\frac{(R^B - 1) + \lambda}{\Phi_A \Phi_B (R^K - 1)} \right)^{1/(\Phi_B - 1)} \tag{5}$$

Intertemporal Euler equation. The first-order condition for n' is:

$$1 = (1 - \mathcal{H}(X))\, \mathbb{E}\left[q(X, X')\, J_1(n'; X')\right] + \mathcal{H}(X)\,(1 - \varphi)\, \mathbb{E}\left[q(X, X')\, J_1\big((1 - \varphi)n'; X'\big)\right] \tag{6}$$

The $(1 - \varphi)$ factor on the second branch arises from the chain rule applied to $J((1 - \varphi)n')$.

Envelope condition. Taking the derivative of $J(n; X)$ with respect to n:

$$J_1(n; X) = \frac{\partial \pi}{\partial n} + 1 + \lambda(1 + \phi_1) = (R^B(X) - 1) + 1 + \lambda(1 + \phi_1) = R^B(X) + \lambda(1 + \phi_1) \tag{7}$$

The term $(R^B - 1)$ is the direct effect of n on deposit costs (the indirect effect through l^* vanishes at the optimum by the lending FOC). The term $+1$ comes from $\partial(n)/\partial n = 1$ in the dividend expression $\pi + n - n'$. The term $\lambda(1 + \phi_1)$ reflects the shadow value of relaxing the leverage constraint: increasing n by one unit relaxes the constraint by $(1 + \phi_1)$ units (since the constraint is $l - n \leq \phi_0 + \phi_1 n$, i.e., $l \leq \phi_0 + (1 + \phi_1)n$).

B.2 Firm's problem

The firm maximizes:

$$\max_{K, H^D} \; AK^\alpha (H^D)^{1-\alpha} - (R^K + \delta - 1)K - w(X)H^D \tag{8}$$

With $\delta = 1$, the first-order conditions are:

$$R^K = \alpha A K^{\alpha - 1} (H^D)^{1-\alpha} \tag{9}$$

$$w(X) = (1 - \alpha) A K^\alpha (H^D)^{-\alpha} \tag{10}$$

B.3 Household's problem

The household maximizes:

$$v(a; X) = \max_{a', B, H^S} \; u(C, H^S) + \beta\, \mathbb{E}\, v(a'; X') \tag{11}$$

subject to:

$$C + p'(X)a' = p(X)a + (R^B(X) - 1)B + w(X)H^S - \xi(B) \tag{12}$$

where $\xi(B) = \zeta_1 B + \zeta_2 B^2$.

With GHH preferences $u = \log\big(C - \eta(H^S)^{1+1/\chi}/(1+1/\chi)\big)$:

Labor supply.

$$H^S = \left(\frac{w(X)}{\eta}\right)^{\chi} \tag{13}$$

Deposit rate. The FOC for B gives:

$$R^B(X) = 1 + \xi'(B) = 1 + \zeta_1 + 2\zeta_2 B \tag{14}$$

Stochastic discount factor. The Euler equation for equity holdings gives:

$$q(X, X') = \beta\,\frac{u_c(C', H^{S'})}{u_c(C, H^S)} \tag{15}$$

With log-GHH utility ($\gamma = 1$), $u_c = 1/\tilde{C}$ where $\tilde{C} = C - \eta(H^S)^{1+1/\chi}/(1+1/\chi)$, so:

$$q(X, X') = \beta\,\frac{\tilde{C}(X)}{\tilde{C}(X')} \tag{16}$$

with current-period composite consumption in the numerator and next-period in the denominator.

Labor market equilibrium. Substituting labor demand $w = (1-\alpha)AK^{\alpha}H^{-\alpha}$ into labor supply $H = (w/\eta)^{\chi}$ and solving:

$$H = \left(\frac{(1-\alpha)AK^{\alpha}}{\eta}\right)^{1/(1/\chi+\alpha)} \tag{17}$$

B.4 Equilibrium conditions

Market clearing.

$$\text{Lending:}\quad L(X) = l(N; X) \tag{18}$$

$$\text{Capital:}\quad \Phi(L(X)) = K(X) \tag{19}$$

$$\text{Deposits:}\quad B = b(N; X) = L - N \tag{20}$$

$$\text{Equity:}\quad a = 1 \tag{21}$$

$$\text{Labor:}\quad H^S = H^D =: H \tag{22}$$

$$\text{Asset price:}\quad p(X) = J(N; X) \tag{23}$$

Aggregate equity transition.

$$N'(X) = \begin{cases} n'(N;X) & \text{with prob. } 1-\mathcal{H}(X) \\ (1-\varphi)\,n'(N;X) & \text{with prob. } \mathcal{H}(X) \end{cases} \tag{24}$$

National accounting identity.

$$Y = C + \xi(B) + K + (N' - N) \tag{25}$$

When a bank failure occurs, a fraction φ of bank equity is destroyed and is *not* rebated to households; the loss is absorbed through the lower realized equity N' (the failure branch of the equity transition above), so bank failures destroy aggregate resources.

Derivation of the national accounting identity. Under competitive factor markets with CRS and $\delta = 1$: $Y = R^K K + wH$ (factor payments exhaust output). From the household budget constraint, aggregate consumption is:

$$C = wH + (R^B - 1)B + D - \xi(B)$$

where $D = \pi + N - N'$ is the aggregate dividend and bank profit is $\pi = (R^K - 1)K - (R^B - 1)B$. Substituting:

$$\begin{aligned} C &= wH + (R^K - 1)K + N - N' - \xi(B) \\ &= (Y - K) + N - N' - \xi(B) \end{aligned}$$

Rearranging yields the identity (25):

$$Y = C + \xi(B) + K + (N' - N).$$

B.5 Social planner's problem

The constrained social planner maximizes the representative household's lifetime utility:

$$\max_{\{n'_t, l_t\}_{t\geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t\, u(C_t, H_t) \tag{26}$$

subject to: (i) the intermediation technology $K_t = \Phi(l_t)$; (ii) the leverage constraint $l_t - n_t \leq \phi_0 + \phi_1 n_t$; (iii) the factor markets (equations (9)–(10)); (iv) the hazard function $\mathcal{H}_t = \sigma(\kappa_1 N_t / L_t + \kappa_0)$; (v) the aggregate equity transition; and (vi) the resource constraint. Crucially, the planner recognizes that the lending choice l_t affects the current-period hazard through $L_t = l_t$ in equilibrium, and that the equity choice n'_t affects next-period hazard through N_{t+1}.

The key difference from the competitive equilibrium is that the planner's intratemporal lending FOC internalizes $\partial\mathcal{H}/\partial l \neq 0$. Since $L = N + B$ and $B = l - n$, choosing more lending raises deposits, increasing leverage and the hazard rate. This generates a Pigouvian wedge $\tau(X)$ that appears in both the lending FOC (intratemporal margin) and the modified return (intertemporal margin).

Modified Euler equation. The planner's Euler equation adds a Pigouvian wedge:

$$1 = (1-\mathcal{H})\,\mathbb{E}\left[q\,J_1^{\text{SPP}}(n';X')\right] + \mathcal{H}\,(1-\varphi)\,\mathbb{E}\left[q\,J_1^{\text{SPP}}\big((1-\varphi)n';X'\big)\right] \tag{27}$$

where:

$$J_1^{\text{SPP}}(n;X) = R^B(X) + \lambda(1+\phi_1) + \tau(X) \tag{28}$$

Pigouvian tax.

$$\tau(X) = \frac{\beta}{u_c(C,H^S)} \cdot \frac{\partial\mathcal{H}}{\partial B} \cdot \mathbb{E}\left[V(n';X') - V\big((1-\varphi)n';X'\big)\right] \tag{29}$$

The derivative of the hazard with respect to deposits is:

$$\frac{\partial\mathcal{H}}{\partial B} = \kappa_1 \cdot \mathcal{H}(1-\mathcal{H}) \cdot \frac{\partial(N/L)}{\partial B} = \kappa_1 \cdot \mathcal{H}(1-\mathcal{H}) \cdot \left(-\frac{N}{L^2}\right) \tag{30}$$

Since $\kappa_1 < 0$ and $-N/L^2 < 0$, we have $\partial\mathcal{H}/\partial B > 0$: higher deposits (more leverage) increase the failure hazard. Since $V(n') > V((1-\varphi)n')$, the Pigouvian tax is positive ($\tau > 0$), inducing the planner to choose lower leverage.

Modified lending FOC.

$$\Phi'(l)(R^K - 1) - (R^B - 1) = \lambda + \tau \tag{31}$$

The positive τ acts as an additional cost of leverage, reducing the planner's lending relative to the decentralized equilibrium.

C Proofs

Proof of Proposition 1. The Pigouvian wedge is $\tau(X) = (\beta/u_c) \cdot (\partial\mathcal{H}/\partial B) \cdot \mathbb{E}[V(n';X') - V((1-\varphi)n';X')]$. Each factor is signed as follows. (i) $\beta/u_c > 0$. (ii) $\partial\mathcal{H}/\partial B = \kappa_1\mathcal{H}(1-\mathcal{H})(-N/L^2) > 0$, since $\kappa_1 < 0$ and $-N/L^2 < 0$, so the product of two negatives is positive. (iii) $V(n';X') - V((1-\varphi)n';X') > 0$ for $\varphi > 0$, since more equity is welfare-improving (the household benefits from higher bank franchise value and lower future hazard rates). Therefore $\tau > 0$. The higher return

$J_1^{\text{SPP}} = R^B + \lambda(1+\phi_1) + \tau > J_1^{\text{CE}} = R^B + \lambda(1+\phi_1)$ implies that the planner's Euler equation requires higher expected marginal value, achieved by retaining more equity. ■

Proof of Proposition 2. The conditional steady state under frozen regime A is a fixed point of $n'(\cdot; A)$. By assumption, $n'(N; A) > N$ for small N (precautionary savings and high marginal returns push equity upward from low levels) and $n'(N; A) < N$ for large N (dividend extraction eventually dominates). Continuity of $n'(\cdot; A)$ and the intermediate value theorem guarantee existence of a fixed point N_A^{CS} satisfying $n'(N_A^{\text{CS}}; A) = N_A^{\text{CS}}$. If $n'(\cdot; A)$ crosses the 45-degree line exactly once, the fixed point is unique. Global stability follows: for any $N_0 < N_A^{\text{CS}}$, $n'(N_0; A) > N_0$ so the forward orbit $\{N_t\}$ is increasing and bounded above by N_A^{CS}; symmetrically for $N_0 > N_A^{\text{CS}}$. ■

Proof of Proposition 3. By Proposition 1, $n'^{\text{SPP}}(N; A) > n'^{\text{CE}}(N; A)$ for all N. The SPP policy function lies strictly above the CE policy function. Since both are continuous and cross the 45-degree line exactly once (by Proposition 2), the SPP crossing occurs at a strictly higher N: $N_A^{\text{CS,SPP}} > N_A^{\text{CS,CE}}$. ■

D Steady state characterization

The deterministic steady state is computed under the assumption of zero failure probability ($\mathcal{H} = 0$). This serves as the initial guess for the RTM iteration; the stochastic equilibrium then endogenously determines positive hazard rates.

Deposit rate. From the Euler equation with $\mathcal{H} = 0$ and $\lambda = 0$: $1 = \beta R^B$, so $R^B_{\text{ss}} = 1/\beta$.

Deposits. From the deposit rate equation (14):

$$B_{\text{ss}} = \frac{1/\beta - 1 - \zeta_1}{2\zeta_2} \tag{32}$$

Capital. The steady-state capital is determined by equating the firm-implied return on capital (from (9)) with the bank-implied return (from the lending FOC (3) with $\lambda = 0$):

$$\alpha A K^{\alpha-1} H(K)^{1-\alpha} = \frac{R^B - 1}{\Phi'(L(K))} + 1 \tag{33}$$

where $H(K)$ is given by (17) and $L(K)$ is obtained by inverting $K = \Phi_A L^{\Phi_B}$.

Table D.3 reports the deterministic steady-state values under the baseline calibration.

Table D.3: Deterministic Steady State

Variable	Symbol	Value
Output	Y	0.3340
Consumption	C	0.2267
Capital	K	0.1064
Labor	H	0.5867
Bank equity	N	0.0413
Deposits	B	0.0417
Loans	L	0.0829
Capital ratio	N/L	0.4975
Return on capital	R^K	1.0361
Deposit rate	R^B	1.0417
Bank profit	π	0.0021
Wage	w	0.3814
Franchise value	J	0.0526
Hazard rate	$\mathcal{H}$	0.0501

Notes: Deterministic steady state with $\mathcal{H} = 0$ and $\lambda = 0$.

E Sensitivity analysis

Table E.4 reports the sensitivity of key model outcomes to ±25% perturbations of five parameters: the hazard slope κ_1, the distress recovery loss φ, the intermediation curvature Φ_B, the leverage slope ϕ_1, and the deposit cost ζ_2. For each perturbation, the model is re-solved from scratch (steady state plus full RTM iteration), and we report the steady-state capital ratio N_{ss}/L_{ss}, the mean ergodic hazard rate $\bar{\mathcal{H}}$, the fraction of periods where the leverage constraint binds, output volatility $\sigma(Y)$, and mean bank equity $\bar{N}$.

Table E.4: Sensitivity Analysis

Parameter	Value	N_{ss}/L_{ss}	$\bar{\mathcal{H}}$	Binding (%)	$\sigma(Y)$	$\bar{N}$
κ_1	-5.1450	0.4975	0.1836	31.9	0.0382	0.0293
κ_1	$-6.8600^{\dagger}$	0.4975	0.0887	17.8	0.0311	0.0334
κ_1	-8.5750	0.4975	0.0373	9.5	0.0256	0.0374
φ	0.4500	0.4975	0.0895	18.2	0.0253	0.0334
φ	$0.6000^{\dagger}$	0.4975	0.0887	17.8	0.0311	0.0334
φ	0.7500	0.4975	0.0993	25.2	0.0408	0.0316
Φ_B	0.6750		*failed to converge*			
Φ_B	$0.9000^{\dagger}$	0.4975	0.0887	17.8	0.0311	0.0334
Φ_B	1.1250	0.6886	0.0344	0.0	0.2551	0.2933
ϕ_1	1.1250	0.4975	0.0658	19.8	0.0298	0.0372
ϕ_1	$1.5000^{\dagger}$	0.4975	0.0887	17.8	0.0311	0.0334
ϕ_1	1.8750	0.4975	0.1117	19.3	0.0316	0.0302
ζ_2	0.3750	0.3300	0.1107	31.6	0.0356	0.0295
ζ_2	$0.5000^{\dagger}$	0.4975	0.0887	17.8	0.0311	0.0334
ζ_2	0.6250	0.5980	0.0645	11.6	0.0275	0.0388

Notes: Each parameter is varied by $\pm 25\%$ around its baseline (†). $\bar{\mathcal{H}}$: mean ergodic hazard rate. Binding: fraction of periods where the leverage constraint binds. $\sigma(Y)$: output volatility. $\bar{N}$: mean bank equity.

F Computational details

F.1 RTM algorithm

The Repeated Transition Method (RTM) solves the model by iterating on a simulated equilibrium path of length $T = 5{,}001$ periods (plus BURNIN $= 500$ periods discarded from the beginning). Each iteration consists of:

1. **Backward step (Euler inversion):** For each period t and TFP state A_t, find the bank's optimal equity choice n'_t by inverting the Euler equation (6). State-contingent expectations are computed using iso-shock matching: for each future TFP state A', identify periods on the current path where $A = A'$ and interpolate allocations at the target equity level n'_t (or $(1-\varphi)n'_t$ for the run branch).

2. **Forward step (simulation):** Given the policy functions from the backward step, simulate the economy forward. The forward step determines: (a) equity choice n'_t from the national accounting identity; (b) run realization from Bernoulli$(\mathcal{H}_t)$ using fixed uniform draws; (c) factual equity N_{t+1} from the run/no-run outcome.

3. **Damping:** Update allocations as $x^{\text{new}} = \omega x^{\text{old}} + (1-\omega)x^{\text{forward}}$ with $\omega = 0.999$ for equity and $\omega = 0.99$ for consumption, deposits, and the constraint multiplier.

4. **Convergence:** The algorithm terminates when $\text{MSE}(x^{\text{old}}, x^{\text{forward}}) < 10^{-8}$.

F.2 GIRF computation

The GIRF is computed by forward-simulating the economy from a given initial state (N_0, A_0) using the converged policy functions, averaged over 1,000 Monte Carlo paths. The initial equity N_0 is set to the 10th or 90th percentile of the ergodic distribution of N.

For the TFP shock GIRF, the shocked path starts in recession ($A_0 = A_L$) and the no-shock path starts in expansion ($A_0 = A_H$). From period 1 onward, both paths draw independently from the Markov transition matrix Π_A.

For the hazard shock GIRF, both paths share the same realized TFP sequence. The shocked path forces a bank failure at $t = 1$; the no-shock path forces no failure. From $t \geq 2$, both paths use shared uniform draws for run realizations, ensuring that differences in hazard outcomes arise solely from the endogenous difference in equity levels.